ExoActor: Exocentric Video Generation as Generalizable Interactive Humanoid Control

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Research Questions

- How can third-person video generation be used as a unified interface for modeling interaction-rich humanoid behaviors without task-specific data collection?
- Can generated exocentric videos be effectively converted into executable robot motions through motion estimation and tracking?
- How does robot-to-human embodiment transfer affect the quality and executability of generated video demonstrations for humanoid control?

Core Idea

ExoActor addresses the challenge of enabling humanoid robots to perform complex interaction-rich tasks by using third-person video generation as an intermediate representation. The key insight is that pretrained video models contain rich implicit knowledge about human-object interactions that can be leveraged for robot control. The system first converts robot observations to human-like representations (embodiment transfer), generates task execution videos using large video models, then extracts 3D motion from these videos and executes it on the robot using a motion tracking controller. This approach decouples high-level interaction planning from low-level control, enabling zero-shot generalization to new tasks without collecting robot-specific data.

Key Formula

Action Chain Decomposition

$$C = \{a_1, a_2, \ldots, a_T\}$$

Decomposes high-level task instruction G into temporally ordered action chain C where each a_t represents an atomic, visually observable, and physically executable action. This structured representation enables more controllable video generation.

Motion Representation

$$M = \{q_t, p_t\}_{t=1}^{T}$$

Represents estimated motion using SMPL parameters where q_t are joint rotations and p_t are global positions at time t. This serves as the interface between video generation and robot execution.

Key Figure

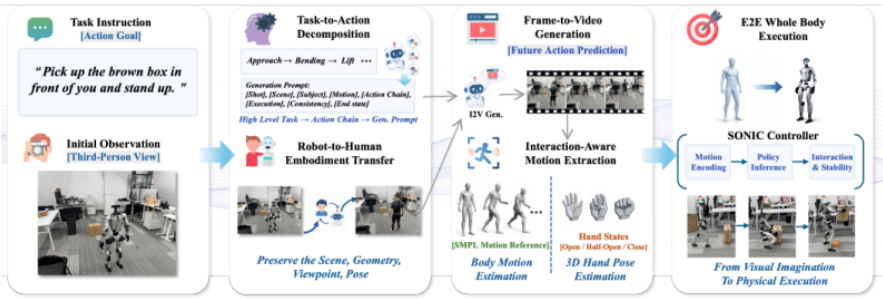

Figure 2: Overview of the ExoActor framework. Given a task instruction and initial observation, the system generates third-person action videos via embodiment transfer and action decomposition, extracts structured motion, and executes it with a motion tracking controller, enabling a transition from visual imagination to physical execution.

Figure 2: Shows the complete ExoActor pipeline from task instruction to robot execution, illustrating the three key stages: video generation with embodiment transfer, motion estimation, and execution via SONIC controller.

Limitations

- Video generation models produce visually plausible but not necessarily physically realizable motions, leading to execution failures
- Motion estimation struggles with fine-grained details like wrist orientation and hand poses, especially under occlusion
- System follows offline pipeline with high latency (cumulative processing time dominates execution)
- Requires small supporting bases under target objects to compensate for height discrepancies in motion estimation